

AI Engineer

Step 4: Build AI Projects

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

An AI project should demonstrate the entire lifecycle: problem definition, data acquisition, preprocessing, baseline, training, evaluation, error analysis and usable inference.

Start with a measurable objective. A simple baseline is valuable because it tells you whether a complex model actually improves the result. Keep experiments reproducible by recording dataset versions, preprocessing, hyperparameters and model artifacts.

Core Concepts — Detailed Breakdown

- Problem definition and target variable.
- Data quality: missing values, duplicates, label errors, imbalance and leakage.
- Baseline: simple rule, statistical model or simple ML model.
- Experiment tracking: dataset, code version, configuration and metrics.
- Evaluation: overall metrics plus slices/error analysis.
- Inference: load the trained artifact and reproduce preprocessing.
- Reproducibility: seeds help but do not guarantee identical results across every environment.
- Documentation: assumptions, limitations, ethical/safety concerns and intended use.

What You Should Be Able To Do

- Define a measurable metric.
- Create a baseline.
- Keep training/inference code separate.
- Save model and preprocessing artifacts.
- Write a failure analysis.
- Document limitations.

Hands-on Project / Practice

- Build one end-to-end project from a public dataset. Provide a training script/notebook, evaluation report, inference script/API and README. Include at least one experiment comparing a baseline with a stronger model.

Recommended References

- Kaggle — <https://www.kaggle.com/>
- Scikit-learn User Guide — https://scikit-learn.org/stable/user_guide.html
- TensorFlow Learn — <https://www.tensorflow.org/learn>

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.